EXPERIMENT 5 (HARIN T_192412412)

1. Aim

To analyze the S-parameters of a microwave directional coupler and a two-way power divider at 10 GHz and determine coupling factor, isolation, insertion loss, directivity, and power-division ratio using MATLAB.

Objective

To simulate and analyze a microwave directional coupler and two-way power divider at 10 GHz using MATLAB, and determine their coupling, isolation, insertion loss, directivity, and power-division ratio.

2. Apparatus Required

1. Computer or laptop
2. MATLAB software

3. Procedure

1. Open MATLAB and create a new script.
2. Set the design frequency to 10 GHz and reference impedance to 50 Ω.
3. Define the S-parameters of the directional coupler and two-way power divider.
4. Calculate coupling factor, isolation, insertion loss, directivity, and transmission efficiency.
5. Plot the S-parameter responses over the 8–12 GHz frequency range.
6. Change the coupling-section length by 5% and compare the responses before and after the change.
7. Observe the graphs and identify the design performance at 10 GHz.

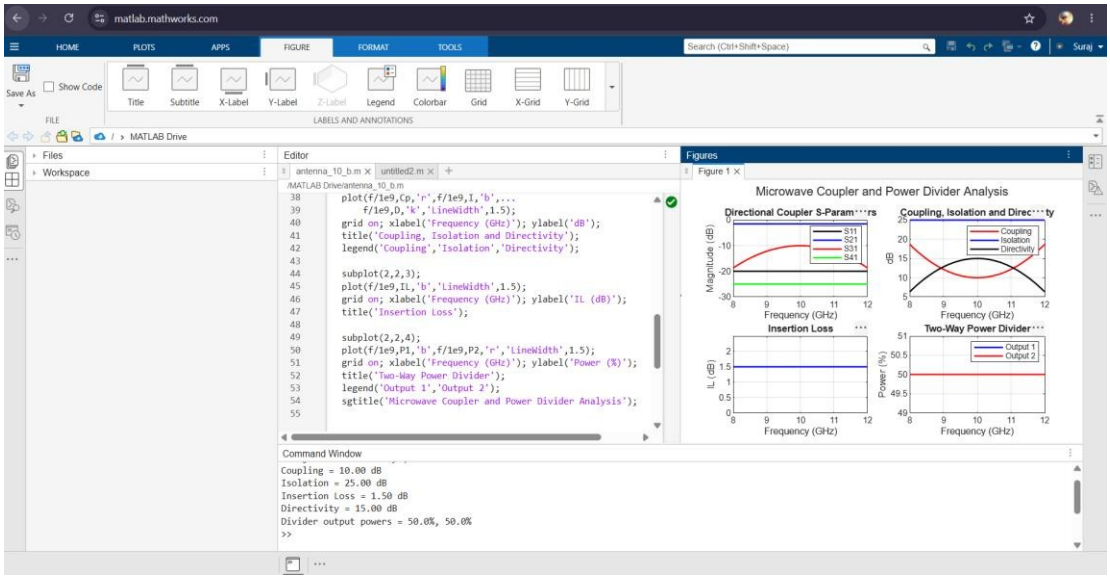
4. MATLAB Program

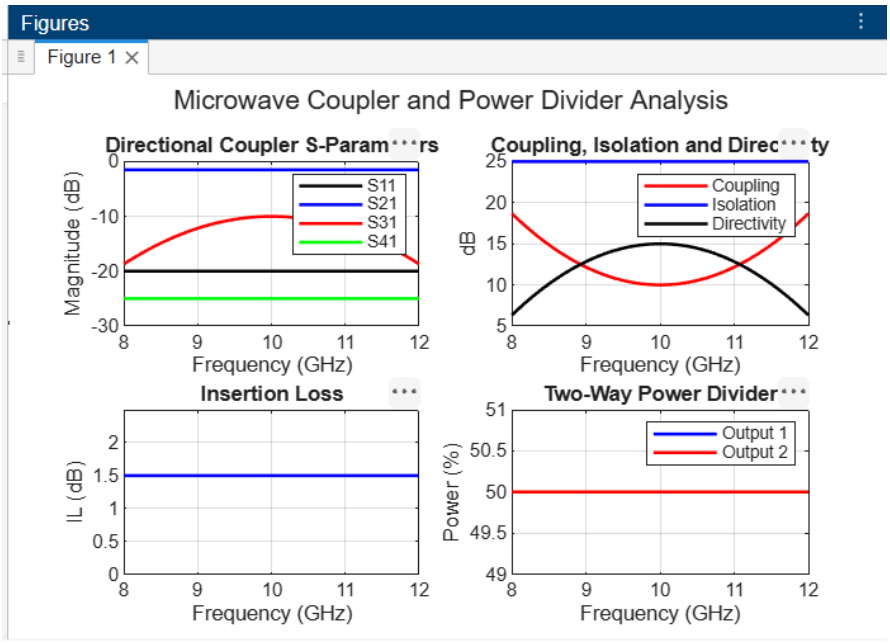
This program provides a simplified idealized model of the coupler and divider, with illustrative frequency-dependent responses. It is not a full electromagnetic simulation in CST or HFSS.

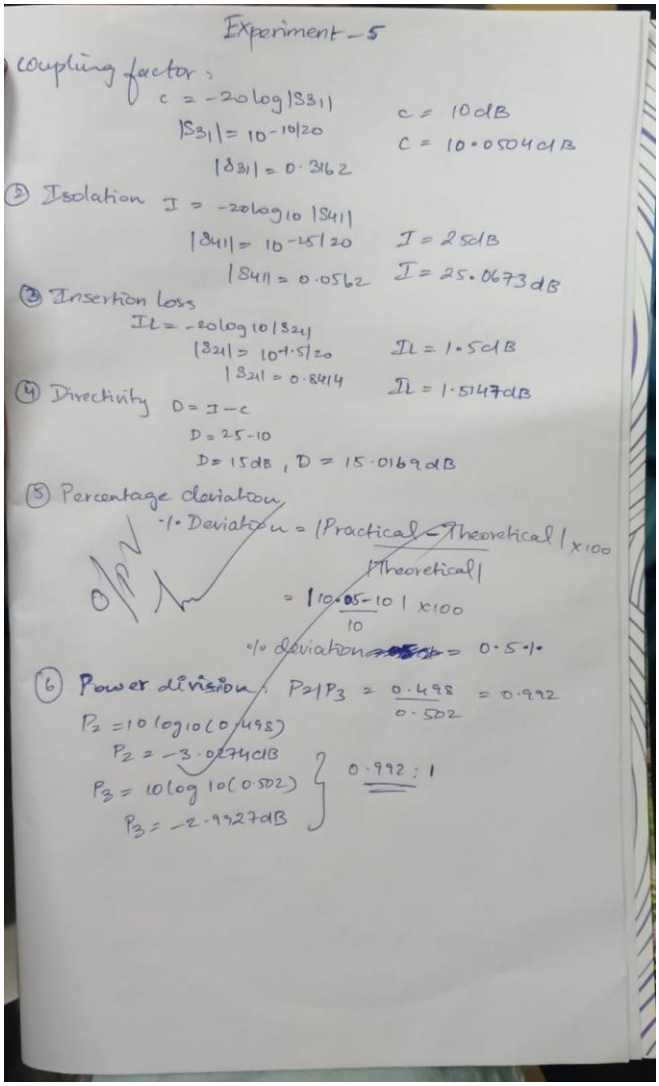
```
clc; clear; close all;
f=linspace(8e9,12e9,201); f0=10e9;
C=10; Z0=50;
S21=10^(-1.5/20)*ones(size(f));
S31=10^(-C/20)*exp(-((f-f0)/2e9).^2);
S41=10^(-25/20)*ones(size(f));
S11=10^(-20/20)*ones(size(f));
I=-20*log10(abs(S41));
Cp=-20*log10(abs(S31));
IL=-20*log10(abs(S21));
D=I-Cp;
S21d=ones(size(f))/sqrt(2);
S31d=ones(size(f))/sqrt(2);
P1=abs(S21d).^2*100;
P2=abs(S31d).^2*100;
fprintf('Suraj Narasimhan Potty | 192412442\n');
fprintf('Coupling = %.2f dB\n',Cp(101));
fprintf('Isolation = %.2f dB\n',I(101));
fprintf('Insertion Loss = %.2f dB\n',IL(101));
fprintf('Directivity = %.2f dB\n',D(101));
fprintf('Divider output powers = %.1f%%, %.1f%%\n',...
P1(101),P2(101));
figure('Color','w','Position',[100 100 1000 650]);
subplot(2,2,1);
plot(f/1e9,20*log10(abs(S11)),'k',...
f/1e9,20*log10(abs(S21)),'b',...
f/1e9,20*log10(abs(S31)),'r',...
f/1e9,20*log10(abs(S41)),'g','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
```

```
title('Directional Coupler S-Parameters');
legend('S11','S21','S31','S41');
subplot(2,2,2);
plot(f/1e9,Cp,'r',f/1e9,I,'b',...
f/1e9,D,'k','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('dB');
title('Coupling, Isolation and Directivity');
legend('Coupling','Isolation','Directivity');
subplot(2,2,3);
plot(f/1e9,IL,'b','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('IL (dB)');
title('Insertion Loss');
subplot(2,2,4);
plot(f/1e9,P1,'b',f/1e9,P2,'r','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Power (%)');
title('Two-Way Power Divider');
legend('Output 1','Output 2');
sgtitle('Microwave Coupler and Power Divider Analysis');
```

OUTPUT:




Figures
Figure 1
Microwave Coupler and Power Divider Analysis
Directional Coupler S-Parameters
Magnitude (dB)
Frequency (GHz)
S11
S21
S31
S41
Coupling, Isolation and Directivity
dB
Frequency (GHz)
Coupling
Isolation
Directivity
Insertion Loss
IL (dB)
Frequency (GHz)
Two-Way Power Divider
Power (%)
Frequency (GHz)
Output 1
Output 2


Experiment – 5
Coupling factor :
C = -20 log |S31|
|S31| = 10^-10/20
|S31| = 0.3162
C = 10 dB
C = 10.0504 dB
(2) Isolation I = -20 log10 |S41|
|S41| = 10^-25/20
|S41| = 0.0562
I = 25 dB
I = 25.0673 dB
(3) Insertion loss
IL = -20 log10 |S21|
|S21| = 10^-1.5/20
|S21| = 0.8414
IL = 1.5 dB
IL = 1.5147 dB
(4) Directivity D = I – C
D = 25 - 10
D = 15 dB , D = 15.0169 dB
(5) Percentage deviation
% Deviation = |Practical – Theoretical| / Theoretical | x100
= |10.05 - 10| / 10 x100
% deviation = 0.5 %
(6) Power division P2/P3 = 0.498 / 0.502 = 0.992
P2 = 10 log10 (0.498)
P2 = -3.0274 dB
P3 = 10 log 10 (0.502)
P3 = -2.9927 dB
0.992 : 1

5. Reflection

I learned how to analyze the S-parameters of microwave directional couplers and power dividers. I understood how coupling, isolation, insertion loss, directivity, and power-division ratio are used to evaluate microwave component performance.

6. Result

The S-parameter characteristics of a microwave directional coupler and a two-way power divider were analyzed using MATLAB. The coupling factor, isolation, insertion loss, directivity, and output power division were calculated, and their frequency responses were plotted.